Synoptic Question 25: MHD Marine Propulsion

Reading Passage

Read the following article and then answer the questions that follow.

Magneto-Hydrodynamic (MHD) propulsion is a conceptual method for propelling marine vessels using electric and magnetic fields, without any moving parts like propellers. The basic principle involves passing seawater through a duct within the vessel. Within this duct, a strong magnetic field (B) is applied perpendicular to the direction of water flow, and an electric field (E) is applied perpendicular to both the flow direction and the magnetic field.

Seawater, being conductive due to dissolved salts, allows an electric current (I) to flow between electrodes placed on the sides of the duct. This current flows through the magnetic field, and the charged ions within the water experience a Lorentz force (F_L). According to Fleming's Left-Hand Rule, this force is directed along the duct, pushing the water out of the rear. By Newton's third law, this expulsion of water creates a forward thrust on the vessel.

The force density (f) on the fluid is given by the vector cross product $f = J \times B$, where J is the current density. For mutually perpendicular fields and flow, the magnitude of the total thrust F_{MHD} produced within the active volume V_{duct} of the duct is approximately $F_{MHD} = JBV_{duct}$. The current density J is related to the electric field E and the seawater conductivity σ by Ohm's Law for fluids, $J = \sigma E$.

A major challenge is generating the required powerful magnetic fields, often necessitating superconducting magnets. The vessel also experiences a hydrodynamic drag force F_D which typically increases with speed v, often modelled as $F_D = kv^2$ where k is a constant.

Data:

Data for a conceptual MHD-powered submersible:

Table 1: Data for MHD Submersible

Parameter	**Value**
Total mass of submersible, m	1.20×10^5 kg
Total volume of submersible hull, V_{hull}	118 m^3
Density of seawater, ρ_{sea}	1025 kg m^{-3}
Conductivity of seawater, σ	4.80 S m^{-1}
MHD duct length (active region), L	5.00 m
MHD duct width (electrode separation), w	0.80 m
MHD duct height, h	0.80 m
Magnetic field strength, B	4.00 T
Voltage applied across electrodes, V_{elec}	1600 V
Drag constant, k	1150 kg m^{-1}
Overall energy conversion efficiency (electrical to kinetic), η	45%

Questions

(a) The submersible is neutrally buoyant when fully submerged.

  (i) Calculate the buoyant force acting on the submerged submersible.

  (ii) Show that the mass of the submersible provided in the data is consistent with it being neutrally buoyant.

(b) The MHD drive generates thrust by applying electric and magnetic fields.

  (i) Calculate the electric field strength E between the electrodes.

  (ii) Calculate the current density J in the seawater within the duct.

  (iii) Calculate the total MHD thrust force, F_{MHD}, generated.

(c) The submersible starts from rest and accelerates.

(i) Calculate the initial acceleration of the submersible, assuming the MHD thrust is instantly at its full value.

(ii) Calculate the maximum theoretical speed (v_{max}) the submersible can reach when the drag force equals the MHD thrust.

(d) Consider the energy and power involved when the submersible travels at a constant speed of 8.00 m s^{-1}.

(i) Calculate the hydrodynamic drag force F_D at this speed.

(ii) Calculate the useful propulsive power P_{prop} being delivered to the water.

(iii) Given the overall efficiency η, calculate the total electrical power P_{elec} consumed by the MHD system.

(iv) Using $P = VI$, calculate the total current I drawn from the power source.

(e) The relationship between thrust F_{MHD} and applied voltage V_{elec} is being investigated.

(i) Show that the thrust is directly proportional to the applied voltage, $F_{MHD} \propto V_{elec}$.

(ii) The maximum speed v_{max} occurs when $F_{MHD} = F_D = kv_{max}^2$. Using this, derive a relationship between v_{max} and V_{elec}.

(iii) If the applied voltage were reduced by half, determine the new maximum speed the submersible could reach.

(f) Calculate the work done by the MHD drive against the drag force as the submersible travels a distance of 5.00 km at the constant speed of 8.00 m s^{-1}.

Solutions

(a) (i) Buoyant Force $F_B = \rho_{sea} V_{hull} g = (1025)(118)(9.81) = 1.185 \times 10^6$ N.

(ii) Weight $W = mg = (1.20 \times 10^5)(9.81) = 1.177 \times 10^6$ N.
Since $W \approx F_B$ (within reasonable rounding), the mass is consistent with neutral buoyancy.

(b) (i) $E = \frac{V_{elec}}{w} = \frac{1600}{0.80} = 2000$ V m^{-1}.

(ii) $J = \sigma E = (4.80)(2000) = 9600$ A m^{-2}.

(iii) Active Volume $V_{duct} = L \times w \times h = (5.00)(0.80)(0.80) = 3.20$ m^3.
$F_{MHD} = JBV_{duct} = (9600)(4.00)(3.20) = 122880$ N $\approx 1.23 \times 10^5$ N.

(c) (i) At rest, $v = 0$, so $F_D = 0$. Net force $F_{net} = F_{MHD}$.
$a = \frac{F_{net}}{m} = \frac{122880}{1.20 \times 10^5} = 1.024$ m s^{-2} ≈ 1.02 m s^{-2}.

(ii) Maximum speed occurs when $F_D = F_{MHD}$.
$kv_{max}^2 = F_{MHD}$
$v_{max} = \sqrt{\frac{F_{MHD}}{k}} = \sqrt{\frac{122880}{1150}} = 10.33$ m s^{-1} ≈ 10.3 m s^{-1}.

(d) Constant speed $v = 8.00$ m s^{-1}.

(i) $F_D = kv^2 = (1150)(8.00)^2 = 73600$ N.
(Note: This is less than F_{MHD}, so this speed is achievable).

(ii) At constant speed, Thrust = Drag. The useful force is equal to the drag force being overcome.
$P_{prop} = \text{Force} \times v = F_D \times v = (73600)(8.00) = 588800$ W ≈ 589 kW.

(iii) $\eta = \frac{P_{prop}}{P_{elec}} \implies P_{elec} = \frac{P_{prop}}{\eta}$
$P_{elec} = \frac{588800}{0.45} = 1.308 \times 10^6$ W ≈ 1.31 MW.

(iv) $I = \frac{P_{elec}}{V_{elec}} = \frac{1.308 \times 10^6}{1600} = 817.5$ A ≈ 818 A.
(Note: This calculation uses V_{elec} as the supply voltage, which is reasonable if the power source is matched to the electrode voltage).

(e) (i) $F_{MHD} = JBV_{duct} = (\sigma E)BV_{duct} = \sigma \left(\frac{V_{elec}}{w}\right) BV_{duct}$.
Since σ, w, B, V_{duct} are constants, $F_{MHD} = (\text{constant}) \times V_{elec}$.
Thus, $F_{MHD} \propto V_{elec}$.

(ii) From (i), $F_{MHD} = CV_{elec}$ where $C = \frac{\sigma B V_{duct}}{w}$.
At max speed, $F_{MHD} = kv_{max}^2$.
$CV_{elec} = kv_{max}^2 \implies v_{max}^2 = \frac{C}{k} V_{elec}$.
$v_{max} = \sqrt{\frac{C}{k}} \sqrt{V_{elec}}$.
Relationship: $v_{max} \propto \sqrt{V_{elec}}$.

(iii) Since $v_{max} \propto \sqrt{V_{elec}}$, if V_{elec} is halved, the new voltage $V'_{elec} = V_{elec}/2$.
$\frac{v'_{max}}{v_{max}} = \sqrt{\frac{V'_{elec}}{V_{elec}}} = \sqrt{\frac{V_{elec}/2}{V_{elec}}} = \sqrt{\frac{1}{2}} = \frac{1}{\sqrt{2}} \approx 0.707$.
New max speed $v'_{max} = v_{max} \times 0.707 = (10.33) \times 0.707 = 7.30$ m s^{-1}.

(f) Distance $d = 5.00$ km $= 5000$ m.
Drag force F_D at 8.00 m s^{-1} is 73600 N (from d(i)).
Work done $W = F_D \times d = (73600)(5000) = 3.68 \times 10^8$ J.